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Bacterial assemblages imply methylmercury production at the rice-soil system

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ABSTRACT

The plant microbiota can affect plant health and fitness by promoting methylmercury (MeHg) production in paddy soil. Although most well-known mercury (Hg) methylators are observed in the soil, it remains unclear how rice rhizosphere assemblages alter MeHg production. Here, we used network analyses of microbial diversity to identify bulk soil (BS), rhizosphere (RS) and root bacterial networks during rice development at Hg gradients. Hg gradients greatly impacted the niche-sharing of taxa significantly relating to MeHg/THg, while plant development had little effect. In RS networks, Hg gradients increased the proportion of MeHg-related nodes in total nodes from 37.88% to 45.76%, but plant development enhanced from 48.59% to 50.41%. The module hub and connector in RS networks included taxa positively (Nitrososphaeracea, Vicinamibacteraceae and Oxalobacteraceae) and negatively (Gracilibacteraceae) correlating with MeHg/THg at the blooming stage. In BS networks, Deinococcaceae and Paludibacteraceae were positively related to MeHg/THg, and constituted the connector at the reviving stage and the module hub at the blooming stage. Soil with an Hg concentration of 30 mg kg^{-1} increased the complexity and connectivity of root microbial networks, although microbial community structure in roots was less affected by Hg gradients and plant development. As most frequent connector in root microbial networks, Desulfovibrionaceae did not significantly correlate with MeHg/THg, but was likely to play an important role in the response to Hg stress.

1. Introduction

Owning to its toxicity, mobility, long-range transport in the atmosphere, and methylation in anaerobic environments, mercury (Hg) is a global pollutant of great concern worldwide (Selin, 2018). By bioaccumulation in the food chain, Hg and its derivatives endanger human health by damaging mental and immune systems, inducing lung, kidney and skin diseases, and even causing death at a high level (EFSA Panel on Contaminants in the Food Chain (CONTAM), 2012; Lavoie et al., 2013). Thus, the Minamata Convention on Hg was adopted in 2013. Normally, fish consumption is the main route of human exposure to methylmercury (MeHg) (Lavoie et al., 2013). However, rice ingestion has been found to constitute another prime pathway of MeHg exposure in areas where rice is the staple food of local residents (Feng and Qiu, 2008; Gong et al., 2018). MeHg bioaccumulated in rice is presumed to derive from MeHg formed in paddy soil (Guo et al., 2021; Rothenberg et al., 2014), where the anaerobic environment allows anaerobic microorganisms to dominate MeHg production (Parks et al., 2013). Consequently, elaborate analysis of microbial methylation in paddy soil, a hot spot for MeHg generation, is of great significance for the evaluation and effective control of MeHg pollution in rice as well as the management of Hg pollution.

A two-gene cluster (*hgcA* and *hgcB*) has been identified to be involved in the methylation of Hg in microorganisms (Parks et al., 2013). In this context, microorganisms containing *hgcAB* genes like Deltaproteobacteria, Firmicutes and Euryarchaeota are reported widely (Gilmour et al., 2013). Overall, the presence of these genes has so far been restricted to anaerobic sulfate (SRB) and iron (IRB) reducing bacteria, methanogens

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and syntrophs (Gilmour et al., 2013; Parks et al., 2013; Tang et al., 2020). SRB like *Desulfobulbus propionicus* strain 1pr3 was among the dominant Hg methylators, while also the IRB like *Geobacter* sp. strain CLFeRB had a high Hg methylation ability (Bravo et al., 2018; Fleming et al., 2006; Schaefer et al., 2011). Despite low Hg methylation ability observed in methanogens (Gilmour et al., 2013), Methanomicrobia was considered to be the principal Hg methylators in Lake Periphyton and other habitats (Hamelin et al., 2011; Yu et al., 2013). Actually, methylation capacities of theses microorganisms depended on strains rather than species or genus (Gilmour et al., 2013; Ma et al., 2019a). Till now, there are around 50 strains with Hg methylation ability (Gilmour et al., 2013; Ma et al., 2019a; Podar et al., 2015), which constitutes only a tiny proportion of the enormous number of microorganisms (Arnoldini et al., 2018). These studies on *hgcAB* enable us to better understand the microbial Hg methylation process. However, the importance of microbial interactions for the microbial Hg methylation process is unconsciously weakened, since only limited information on the role of non-Hg methylators in the methylation process is available (Liu et al., 2019).

Microorganisms, including viruses, bacteria, archaea and protists, do not exist as individual groups, but form complex ecological networks, where interactions among members can produce positive and negative effects or no impacts on the species involved (Faust and Raes, 2012), and are vital for community assembly and ecosystem functioning (Fuhrman, 2009; Hallam and McCutcheon, 2015). Moreover, ecological processes determine the selective conditions that drive evolutionary changes conversely feeding back into patterns that affect microbial diversity, thereby boosting further evolutionary changes in iterative and continuous cycles (Hansen, et al., 2007). Apparently, the microbial community is a complex and dynamic system, in which interactions between various types determine the relationship between microbial cells and their biological and abiotic environment (Rainey and Quistad, 2020). In spite of imperfections and uncertainties, dynamic processes within natural microbial communities can be quantified and their quantifications can lead to reliable inferences (Garud et al., 2019). Regrettably, there is little information about microbial dynamic processes involved in methylation in paddy soil that contains most of the identified Hg-methylating microorganisms. Network analysis provides an effective tool to explore the interrelationships of microorganisms and the dynamics and organization of their niches (Faust and Raes, 2012; Ma et al., 2020; Shi et al., 2016), which simultaneously can be used to explore complex microbial assemblies in environments such as humans, oceans, groundwater, and soil (Ma et al., 2020; Shi et al., 2016). Therefore, we investigated the changes of microbial communities in bulk soil, rhizosphere and roots throughout rice development at Hg gradients to understand the links between these changes and MeHg production. With this approach we aimed to identify differences in bacterial assemblages in different niches of paddy ecosystem depending on MeHg production. To further improve knowledge of the response of microbial communities to MeHg production in paddy soil, we constructed networks of MeHg generation at the rice-soil interface by microorganisms classified as Hg-promoted and inhibited methylators on correlating these networks with MeHg/THg (Drott et al., 2008; Liu et al., 2019). We hypothesized that Hg methylation and MeHg demethylation in bulk soil differed from those in rhizosphere, and keystones in bulk soil, rhizosphere and root networks may not be the known Hg methylators.

2. Material and methods

2.1. Soil experiment and sample collection

Soil collected at a rice paddy field in the suburbs of Beibei (106.420016 degrees East and 29.818452 degrees North), Chongqing, China in 2018, was air-dried, mashed, sieved with 20 μm, and then placed into plastic pots (40 kg pot^{-1}) with a length, width and height of 90, 36 and 25 cm respectively. The average THg concentration of the sieved soil was 0.15 mg kg^{-1}. The soil is neutral purple soil developed from purple sand mudstone of Jurassic Shaximiao Formation. Its pH was 7.5, SO_4^{2-} concentration was 22.4 mg kg^{-1}, NH_4^+ concentration was 43.3 mg kg^{-1}, and organic matter content was 9.1 g kg^{-1}. Base fertilizers ($CO(NH_2)_2$, 150 mg kg^{-1}; $Ca(H_2PO_4)_2$, 100 mg kg^{-1}; KCl, 85 mg kg^{-1}) were added into the pots, stirred well, and then the $HgCl_2$ solution was applied to produce an Hg gradient (i.e. 0, 0.3, 3 and 30 mg Hg kg^{-1} soil, which were marked as Hg0, Hg0.3, Hg3 and Hg30 respectively) similar to previous studies (Frossard et al., 2017; Ma et al., 2019b). These concentrations were selected by collecting data from the literature on the effects of Hg pollution and metals on soil microbial communities (Ranjard et al., 2000; Müller et al., 2001). The lowest Hg concentration (0.3 mg Hg kg^{-1} soil) was in the range of concentrations detected in soils without surrounding Hg sources (Rieder et al., 2011). Higher concentrations of Hg (3 mg Hg kg^{-1} soil) were within the concentration range of contaminated soil in the XiuShan YangShikeng Hg mining area in southwestern China (Ma et al., 2019b). The highest concentration used in this study (30 mg Hg g^{-1} soil) caused severe diversity loss and altered microbial community structure and composition (Frossard et al., 2017). Each Hg treatment had three replicates. During three weeks of soil aging, the water surface above the soil was kept at 3–5 cm, which was also the height of the water surface during the experiment. Finally, 60 rice seedlings (Fengyou 210) of similar size were transplanted into the pots and cultivated in the greenhouse. After transplanting, bulk soil (BS), rhizosphere soil (RS) and root samples were collected at 20, 40, 80, 105 and 122 days, which was in line with reviving, tillering, blooming, filling and maturing stages marked as R, T, B, F and M, respectively. For sample collection, the whole rice plants were dug out of soil and gently shaken, with the fallen soil defined as BS and the soil attached to roots as RS. If there was no the fallen soil, we deemed the soil remaining in the pot as BS. The collection time of the root sample was the same time as the rhizosphere and bulk soil samples. After soil samples were collected, rice roots were washed with tap and deionized water to remove remaining soil adhered to the roots. All samples were stored temporarily in liquid nitrogen, transported back to the laboratory within 24 h, and stored at −80 °C for subsequent analyses.

2.2. Hg analysis

Measurement of THg and MeHg concentrations in soil and root samples were performed as previously reported (Guo et al., 2021; Ma et al., 2019b). Soil and root samples were digested by 1:3 (v/v) HCl + HNO_3 and 4:1 (v/v) HNO_3 + H_2SO_4, respectively, and THg contents were determined by gas chromatography-cold vapor atomic fluorescence spectrometry (GC-CVAFS). Soil and root samples were digested with KOH-methanol (250 g L^{-1})/solvent extraction and $CuSO_4$ (2 mol L^{-1})/solvent extraction respectively, to quantify MeHg by GC-CVAFS. Free solvated Hg (II) species are thermodynamically unfavorable in solution (Riccardi et al., 2013), and some chemical forms of Hg (for example Hg-cysteine, Hg-GSH, etc.) were found to be bioavailable for some Hg methylators (*Geobacter sulfurreducens* PCA, *Desulfovibrio desulfuricans* ND132 etc.), may not be bioavailable for other species (Gilmour et al., 2011; Schaefer et al., 2011; Graham et al., 2012). Bioavailable Hg include water extractable Hg (WHg) ($HgCl_2$ and $HgSO_4$) and organic acid soluble Hg ($HgSO_4$, HgO, Hg sorbed to oxides of Fe and Mn), in which WHg has high mobility and organic acid soluble Hg has potential mobility (Bloom et al., 2003; Gamboa-Herrera et al., 2021). The correlation (R^2 = 0.5573) between WHg and MeHg potential is not poor, which was only lower than that between organic Hg (R^2 = 0.7547) and MeHg potential (Bloom et al., 2003; Gamboa-Herrera et al., 2021). Furthermore, there was a negative correlation between organic acid soluble Hg (R^2 = 0.0359) and MeHg potential (Bloom et al., 2003; Gamboa-Herrera et al., 2021). Thus, WHg could be deemed as bioavailable Hg. Measurement of WHg contents in soil was performed as previously reported (Bloom et al., 2003; Gamboa-Herrera et al., 2021). 2 g soil sample and 20 mL reagent-grade double deionized water were added in a 50 mL centrifuge tube, vibrate at room temperature for 18 ±

4 h (30 rpt min^{-1}), centrifuge separation (3000 rpt min^{-1}, 15 min) and filter the supernatant (0.45 μm). WHg content were determined by GC-CVAFS. Details of these methods are described in the Supporting Information.

2.3. DNA extraction, sequencing and analysis, and quantification of functional genes

The genomic DNA was extracted from soil and root samples using the Fast PrepDNA Isolation kit (MP Biomedicals, Carlsbad, CA, USA) according to the instructions. The DNA extracted was checked on 1% agarose gel, while the concentration and purity were determined with a NanoDrop 2000 UV–vis spectrophotometer (Thermo Fisher Scientific, Wilmington, MA, USA). The PCR amplification of the 16S rRNA gene was performed as follows: initial denaturation at 95 °C for 3 min, followed by 27 cycles of denaturation at 95 °C for 30 s, annealing at 55 °C for 30 s and extension at 72 °C for 45 s, single extension at 72 °C for 10 min, and ending at 4 °C. The PCR mixtures contained 5 × TransStart FastPfu buffer 4 μL, 2.5 mM dNTPs 2 μL, forward primer (5 μM) 0.8 μL, reverse primer (5 μM) 0.8 μL, TransStart FastPfu DNA Polymerase 0.4 μL, template DNA 10 ng, and finally ddH_2O up to 20 μL. PCR reactions were performed in triplicate. The PCR product was extracted from 2% agarose gel and purified using the AxyPrep DNA Gel Extraction Kit (Axygen Biosciences, Union City, CA, USA) according to the manufacturer's instructions and quantified using a Quantus™ Fluorometer (Promega, USA). Purified amplicons were pooled in equimolar and paired-end sequenced on an Illumina MiSeq PE300 platform (Illumina, San Diego, USA) according to standard protocols by Majorbio Bio-Pharm Technology Co. Ltd. (Shanghai, China). Absolute quantifications of key functional genes (*drsB*, *merA*, *hgcA* and *mcrA*) associated with Hg transformation were conducted by q-PCR with an ABI3500 q-PCR system (Thermo Fisher Scientific). The primers used for q-PCR are shown in Table S1. The hypervariable region V4 of the bacterial 16S rRNA gene was amplified with primer pairs 515F (5′-GTGCCAGCMGCCGCGGTAA-3′) and 806R (5′-GGACTACHVGGGTWTCTAAT-3′) by an ABI GeneAmp® 9700 PCR thermocycler (ABI, CA, USA). The parameters of the q-PCR reaction, construction of the standard plasmids and processing of sequencing data are described in the Supporting Information. The sequencing data of this study have been deposited in the Gene Expression Omnibus with accession number PRJNA721025.

The raw 16S rRNA gene sequencing reads were demultiplexed, quality-filtered by fastp version 0.20.0 (Chen et al., 2018) and merged by FLASH version 1.2.7 (Magoč and salzberg, 2011) with the following criteria. (i) the 300 bp reads were truncated at any site receiving an average quality score of <20 over a 50 bp sliding window, and the truncated reads shorter than 50 bp were discarded, reads containing ambiguous characters were also discarded; (ii) only overlapping sequences longer than 10 bp were assembled according to their overlapped sequence. The maximum mismatch ratio of overlap region is 0.2. Reads that could not be assembled were discarded; (iii) Samples were distinguished according to the barcode and primers, and the sequence direction was adjusted, exact barcode matching, 2 nucleotide mismatch in primer matching.

Operational taxonomic units (OTUs) with 97% similarity cutoff (Edgar, 2013; Stackebrandt and Goebel, 1994) were clustered using UPARSE version 7.1 (Edgar, 2013), and chimeric sequences were identified and removed. The taxonomy of each OTU representative sequence was analyzed by RDP Classifier version 2.2 (Wang et al., 2007) against the 16S rRNA database (Silva v138) using confidence threshold of 0.7. At the level of 0.03, the coverage of sample sequencing was about 95.2% and the end of the dilution curve was smooth at the same sequencing depth (Fig. S1), indicating that although the coverage rate was not 100%, the sequencing results were within a credible range to explain the microbial communities in the samples.

2.4. Microbial network analysis

Relationships between species and environmental traits can be predicted by network inference, in which the environmental factor can be treated as additional species via similarity-based approaches (Fuhrman, 2009). Strong relationships between microorganisms and concentrations of nutrients like nitrites and nitrates can identify species involved in global biogeochemical cycling (Fuhrman, 2009), and the correlation between microbes. Plus, metal (loid) concentrations can pinpoint species associated to metal (loid) resistance or metal (loid) transformation (Sun et al., 2018). MeHg/THg can be used as an indicator of Hg methylation potential (Drott et al., 2008; Liu et al., 2019). Therefore, by analyzing the Spearman's correlation between relative abundances of operational taxonomic units (OTUs) and MeHg/THg in BS and RS, we divided OTUs with no, positive and negative correlations with MeHg/THg into non-Hg (N), Hg-promoted (P) and Hg-inhibited (I) methylators respectively, when the *P* value was less than 0.05. OTUs only present in roots were assigned to methylators with no related information (None). To reveal potential interactions between microbes associated with Hg methylation in paddy fields, each node in networks was given the nature of Hg-related functions through the taxa response to MeHg/THg. Construction of microbial networks is shown in the Supporting Information.

3. Results

3.1. Blooming and filling stages are important for MeHg production in soil during rice development

Previous studies reported that both plant development and Hg gradient could affect MeHg production in the rhizosphere of paddy soil (Guo et al., 2021; Ma et al., 2019b). To confirm their influence on Hg methylation in different niches of paddy soil, we investigated the differences in MeHg, MeHg/THg, and WHg contents between RS and BS exposed to 0, 0.3, 3 and 30 mg Hg kg^{-1} soil (Hg0, Hg0.3, Hg3 and Hg30, respectively) (Fig. 1A,B and S2A). Compared with Hg0, Hg3 and Hg30 significantly increased the MeHg concentrations in RS and BS across all developmental stages of rice. Hg3 (BS, $P < 0.001$) enhanced, whereas Hg30 (RS, $P < 0.01$) decreased MeHg/THg in RS and BS. However, when different stages of rice development were analyzed, the alterations in MeHg and MeHg/THg in BS and RS along the Hg gradient were significant only at the blooming and filling stages. MeHg/THg was enhanced in BS ($P < 0.05$) and RS ($P < 0.001$) at Hg0.3 and RS ($P < 0.001$) at Hg30 in the blooming stage. Increased levels of MeHg (RS, $P < 0.001$) and MeHg/THg (BS, $P < 0.001$; RS, $P < 0.001$) were observed in the soil at Hg3 at the filling stage when compared with the reviving stage. In addition to the activity of Hg methylators (Gilmour et al., 2013), the level of Hg methylation generally depends on the amount of bioavailable Hg. Soil WHg content varied along the Hg gradient (Fig. 1B). Although there was no difference in WHg between the soil of Hg0 and Hg0.3 ($P > 0.05$) and Hg3 ($P > 0.05$), Hg30 (BS, $P < 0.001$; RS, $P < 0.01$) enhanced WHg concentrations in both RS and BS. However, independent of the amount of Hg^{2+} exposure, the concentration of WHg in RS was lower than that in BS. This difference was significant at the blooming and filling stages of rice development. The THg concentration in roots at Hg3 and Hg30 was significantly higher than that at Hg0, especially at the filling stage (Fig. S2B). The MeHg concentration in roots decreased gradually with progressing growth of rice (Fig. S2C).

3.2. Hg gradient affects microbial community structure in RS and BS

The microbial community structure of BS, RS and roots was characterized to elucidate the effects of the rice developmental stage on microbial communities in different niches of Hg-polluted paddy fields. We obtained a total of 10,232,661 sequences (range 30370–74839) and identified 9891 bacterial and 120 archaeal operational taxonomic units (OTUs) across 180 samples. Hg3 reduced the microbial α-diversity of RS

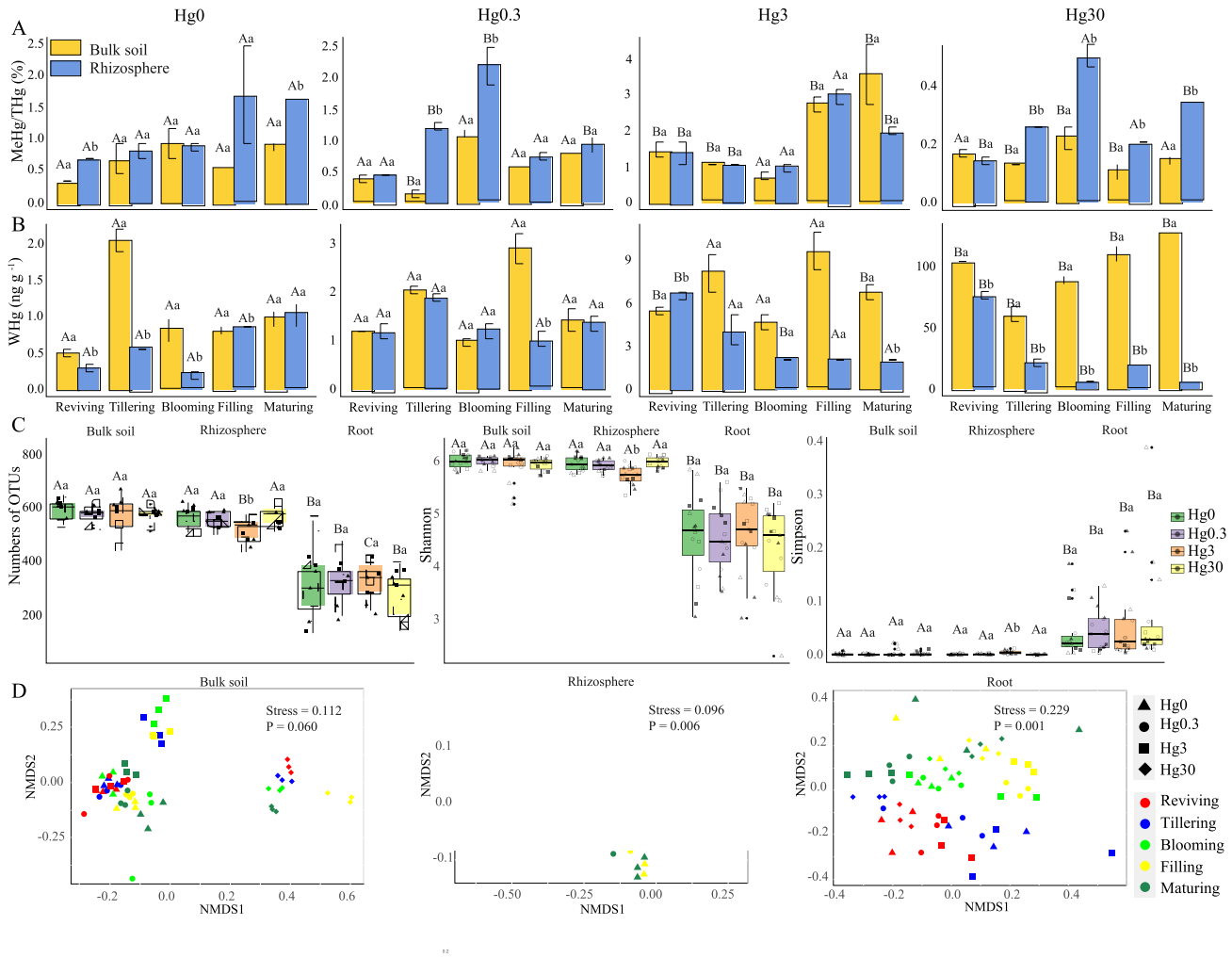


Fig. 1. MeHg/THg (A) and WHg (B) in BS and RS and their microbial α (C) and β (D) diversities: (A and B). Different uppercase letters indicate significant differences between Hg treatments and Hg0 in BS or RS at the same developmental stage of rice; different lowercase letters indicate significant differences between BS and RS. (C). Different uppercase letters indicate significant differences among BS, RS, and roots at the same developmental stage of rice; different lowercase letters indicate significant differences among Hg treatments; hollow circle, hollow triangle, hollow square, solid square, solid triangle and solid circle represent reviving stage, tillering stage, blooming stage, filling stage, maturing stage and outlier, respectively.

based on the number of OTUs (490 ± 39–561 ± 15), Simpson (0.0046 ± 0.001–0.011 ± 0.004) and Shannon (5.54 ± 0.17–5.93 ± 0.06) estimators (Fig. 1C), but had little effect on microbial communities of BS and roots (Fig. 1D). Apparently, the microbial community structure changed in RS at Hg3. In contrast, the microbial community of Hg30 differed from that of other Hg treatments in different niches except for the roots, although Hg30 had little effect on microbial α-diversity. The microbial community and α-diversity of roots were almost unaffected by the Hg gradient, and difference of microbial α-diversity of different niches among the stages of rice development were not observed (Fig. S3). However, the microbial community seemed to differ between vegetative and reproductive growth (Fig. 1D).

The structure of microbial communities in Hg3 and Hg30 treatments differed from Hg0, and between vegetative and reproductive growth (Fig. S4 and S5). Therefore, we focused on dominant microbes significantly influenced by Hg3 and Hg30, as well as the blooming and filling stages. Compared with Hg0, increased relative abundances of Oxalobacteraceae (BS, 10.52%; RS, 11.85%) and Xanthomonadaceae (BS, 3.60%) in the class of Gammaproteobacteria appeared in the soil of Hg3, whereas Hg30 reduced the relative abundances of Oxalobacteraceae (BS, 1.32%; RS, 1.01%), Nitrosomonadaceae (BS, 0.91%; RS, 0.47%) and Xanthomonadaceae (BS, 0.31%) in the class of Gammaproteobacteria, Sphingomonadaceae (BS, 0.61%; RS, 0.44%) in the class of Alphaproteobacteria, Vicinamibacteraceae (BS, 0.68%; RS, 0.60%) in the class of Vicinamibacteria, and Gemmatimonadaceae (BS, 0.92%) in the class of Gemmatimonadetes in the soil (Fig. S6A and S7A). The abundances of these families showed a significantly positive correlation with MeHg/THg in the soil (Fig. S8A). Moreover, the relative abundances of Anaerolineaceae (BS, 5.82%; BS, 8.96%), Bacillaceae (BS, 4.42%; RS, 3.18%), Geobacteraceae (RS, 2.01%) and 67–14 (RS, 5.82%), which had a significant positive correlation with WHg and THg in the soil (Fig. S8A), increased in the Hg30 treatment (Fig. S7A). Nitrososphaeraceae, belonging to the Crenarchaeota phylum, did not correlate with MeHg/THg in the soil. At the blooming and filling stages, the relative abundances of Chitinophagaceae (B, 2.41%; F, 2.47%) and Comamonadaceae (B, 2.01%; F, 1.51%) were reduced, and the relative abundance of Bacillaceae in BS (B, 3.86%; F, 3.21%) was enhanced. While in RS the relative abundances of Chitinophagaceae (B, 1.91%; F, 1.89%) and Sphingomonadaceae (B, 1.16%; F, 1.16%) were reduced and those of Geobacteraceae (B, 2.11%) and Anaerolineaceae (B, 7.65%; F, 8.29%) were increased (Fig. S6B and S7B). However, Hg gradient and plant developmental stage had little effect on microbial community structure of the roots (Fig. S6-S8).

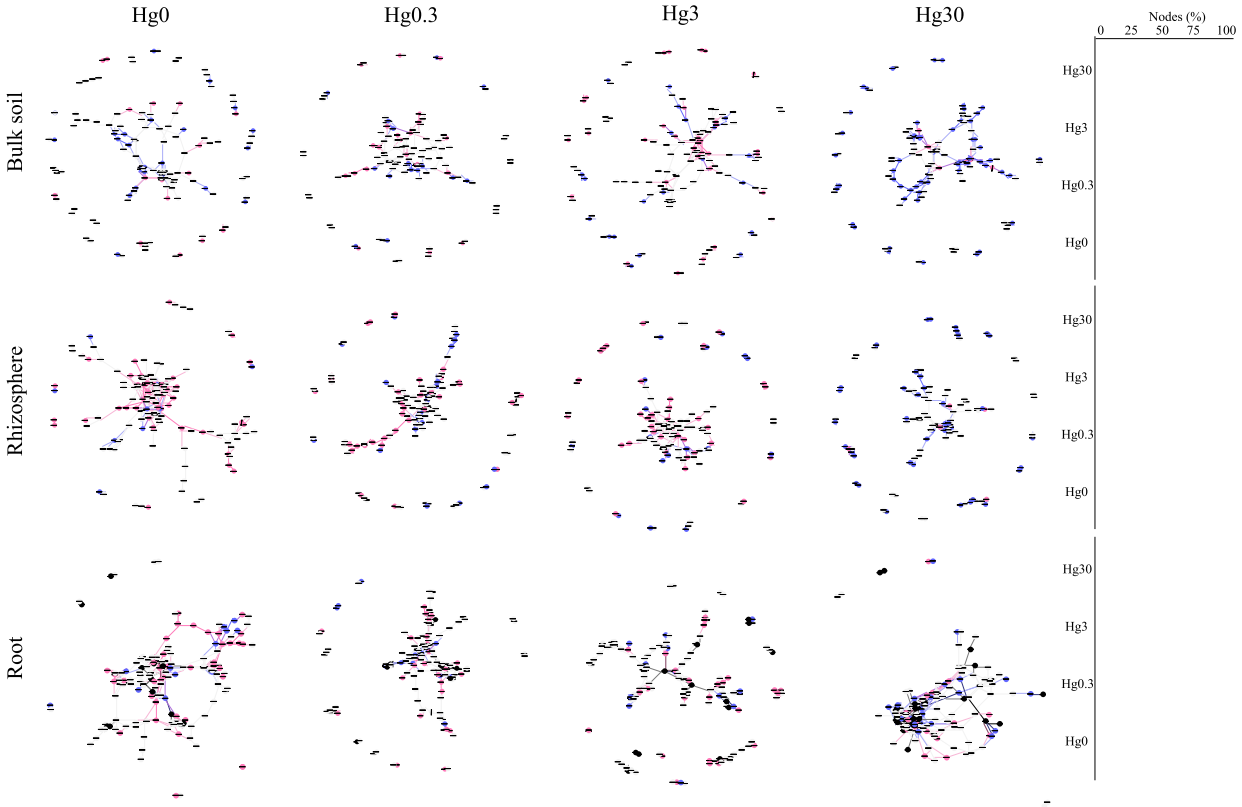


Fig. 2. Microbial networks in BS, RS and roots along an Hg gradient and the proportion of nodes owned by non-Hg (gray), Hg-promoted (red) and Hg-inhibited (blue) methylators in the networks. Black indicates OTUs that only appeared in roots. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.3. Microbial dynamic patterns in MeHg production

To further identify functions of microbes that differ in abundances along the Hg gradient or plant development, clarify the relationship between non-Hg and Hg methylators, and characterize key microbial taxa in the process of Hg methylation, we performed co-occurrence network analyses based on taxa response to MeHg/THg on microbial diversity at the OTU level (Fig. 2 and S9). Root networks were larger than RS and BS networks along the Hg gradient (Table S2) despite the fact that root assemblages contained fewer OTUs than RS and BS assemblages (Fig. 1C). However, along rice plant development, root networks were smaller than RS and BS networks (Table S3). In addition to the root network at Hg30 containing 895 links, the connectivity and complex of BS, RS and root networks showed no differences between Hg treatments, or rice plant development (Fig. S10). The size and connectivity of networks along Hg gradient, which were significantly correlated with Sobs, Shannon and Simpson estimators, increased with the decline in microbial diversity. However, the size and connectivity of networks during rice plant development decreased with the decline in microbial diversity.

We focused on 17 out of 27 networks (Fig. 2 and S9), in line with the spatiotemporal characteristics of the distribution of MeHg (Fig. 1 and S2) that the highest MeHg/THg appeared in Hg3, at blooming and filling stages but lowest in Hg30. Analysis of these networks allows for better identification of MeHg-associated assemblages that potentially interact with or share niches within BS, RS and roots. Among the observed potential interactions, the positive correlation (green lines) accounted for 70.01–94.86% (Figs. 3-5 and S11-S13). The proportion of putative MeHg-associated taxa greatly changed under Hg stress, especially Hg -promoted (6.03%) and -inhibited (50.00%) methylators in the BS network of Hg30 (Fig. 2), while it was less effected by rice plant development (Fig. S9). Hg30 reduced and enhanced the proportion of Hg -promoted (11.18%) and -inhibited (14.91%) methylators in the root network, respectively. Modules in networks along the Hg gradient were dominated by non-Hg methylators, except for modules in the BS network of Hg30 that were mostly dominated by Hg-inhibited methylators (Figs. 3-5). In networks along rice plant development, modules were not dominated by non-Hg methylators except for root networks (Fig. S11-S13).

No network hubs and a total of 9 module hubs and 30 connectors were found in 17 networks (Fig. 6A, D and Table S4). There were 5 non-Hg, 1 Hg-inhibited and 3 Hg-promoted methylators in the module hubs, and 18 connectors were found in the root network of Hg30. Module hubs and connectors are thought to be keystone taxa due to their critical roles in network topology (Shi et al., 2016). Since they accounted for 54.55% of all module hubs and connectors, members of Chloroflexi and Desulfobacterota phyla are the most prominent keystone taxa in the RS networks, which included taxa from the families Anaerolineaceae, Syntrophobacteraceae and Desulfovibrionaceae (Fig. 4 and Table S4). Members of Desulfobacterota, Firmicutes and Proteobacteria were most prominent keystone taxa in root networks, accounting for 60.00% of all module hubs and connectors. Putative keystone taxa include taxa from the families Desulfovibrionaceae, Oscillospiraceae, Sphingomonadaceae and Comamonadaceae (Fig. 5 and Table S4). Although taxa from the same family, namely Anaerolineaceae and Nitrososphaeraceae, did re-occur between different networks, there was a single taxon, OTU14475 assigned to Desulfovibrionaceae that acted as a keystone for different networks, i.e. the RS network of Hg3 and the root network of Hg0 (Table S4). Compared with the networks affected by the Hg gradient, the networks impacted by rice plant development had fewer module hubs and connectors. Either way, the relative abundance of module hubs (0.002–0.13%) and connectors (0.001–0.06%) were low (Fig. 6).

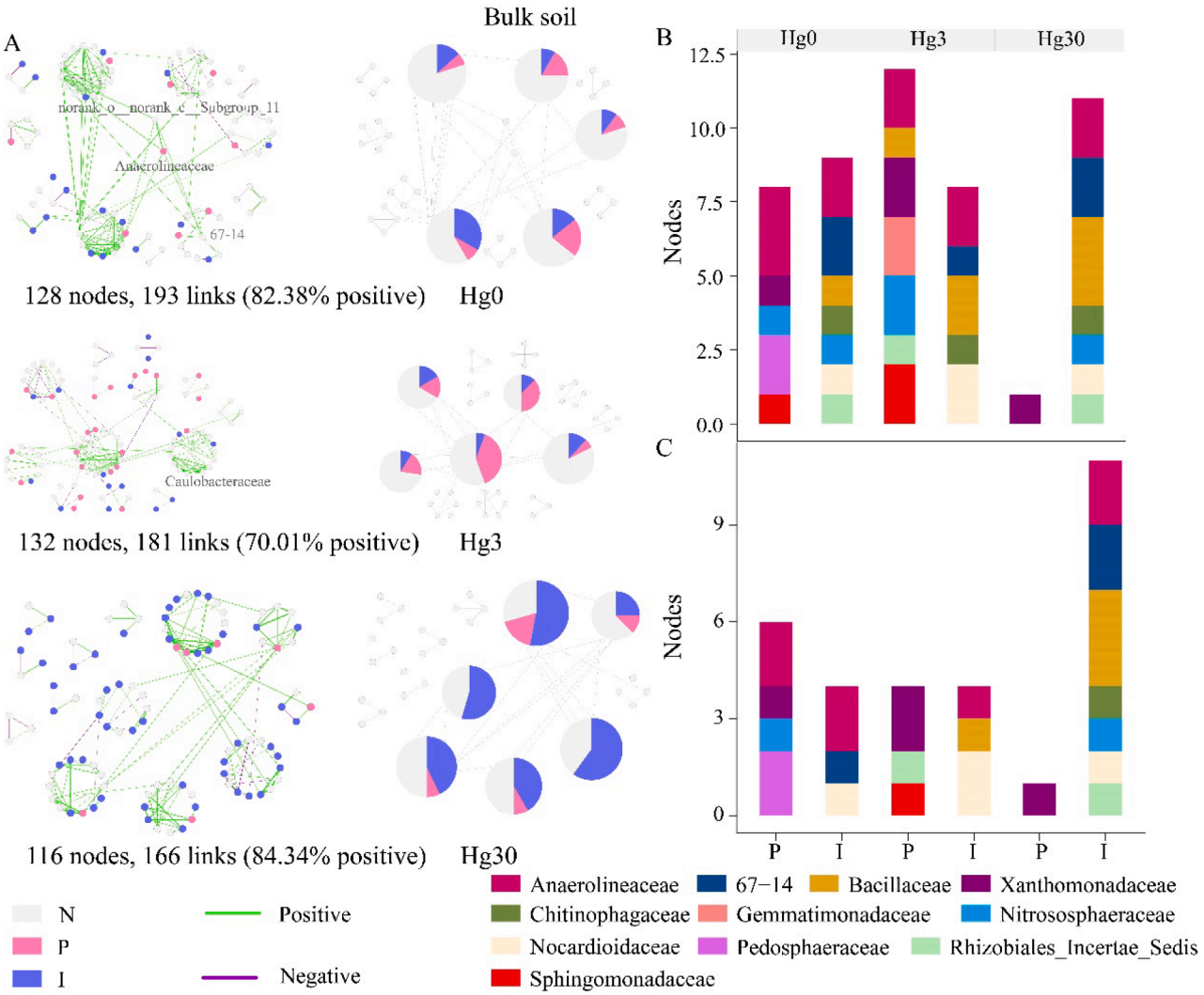


Fig. 3. Highly connected modules within BS networks along an Hg gradient (A) and the number of nodes of the top bacteriaceae in different putative MeHg-associated taxa in the network (B) and highly connected modules (C). Colours of nodes indicate different putative Hg-associated taxa, pie charts represent the composition of modules with >8 nodes, and the nodes labeled with bacteriaceae are module hubs and connectors, among which nodes in the middle of the module are module hubs, and other nodes are connectors.

4. Discussion

Geobacteraceae assigned to Hg-inhibited taxa was found among the bacteria families with the largest number of nodes in RS network of rice at Hg30 (Fig. 4B), suggesting that species like *Geobacter bemidjiensis* could be involved in Hg methylation and demethylation processes (Lu et al., 2016). Bacillaceae belonging to Hg-promoted taxa was found among the bacteria families with the largest number of nodes in BS network of Hg3 (Fig. 3B), indicating that species like *Bacillus megaterium* can participate in the Hg methylation process (Vonk and Sijpesteijn, 1973). These results reflect the reliability of patterns based on taxa responding to MeHg production. To our knowledge, this is the first time that interactions within MeHg-associated microorganisms have been characterized in different ecological niches in the rice-soil interface, thereby providing a new perspective on the dynamics of MeHg-associated microbial communities in rice paddies.

4.1. Different microbial mechanisms of MeHg production under Hg stress between BS and RS

The methylation of Hg is controlled not only by the families of putative Hg-methylating microorganisms in RS, but also by non-Hg methylators in BS. Quantification of *hgcA* in RS showed that 3 mg kg^{-1} Hg concentration increased the abundance of *hgcA* (Fig. S14), which may be related to the relative abundance of Geobacteraceae in RS (Fig. S7A). In paddy fields containing most of the currently known and identified methylators, SRB are considered to be the main Hg-methylating microorganisms (Liu et al., 2014). But recent research indicates that among Hg methylators, the highest abundance is observed in IRB and methanogens, whereas the abundance of SRB is the lowest (Liu et al., 2018). Apparently, methanogens and IRB, rather than SRB, may be dominant methylators in paddy soils. Indeed, the dominant methylators in rice rhizosphere were iron-reducing Geobacteraceae, which had a large number of nodes in RS networks (Fig. 4) and were also the dominant family of microbial communities in RS (Fig. S7A). Despite low abundance (Fig. S7), Desulfovibrionaceae, commonly found in paddy fields, sediments and forest soil (Podar et al., 2015), was likely to be related to Syntrophobacteraceae as driving factor for MeHg production in RS, because both were connectors of RS network (Fig. 4). Although both Desulfovibrionaceae and Syntrophobacteraceae contain Hg-methylating genes (Gilmour et al., 2013; Podar et al., 2015; Tang et al., 2020), the connectors were assigned to be non-Hg methylators. Syntrophs, primarily belonging to Deltaproteobacteria, can obtain energy from syntrophy with SRB or methanogens (Gilmour et al., 2013).

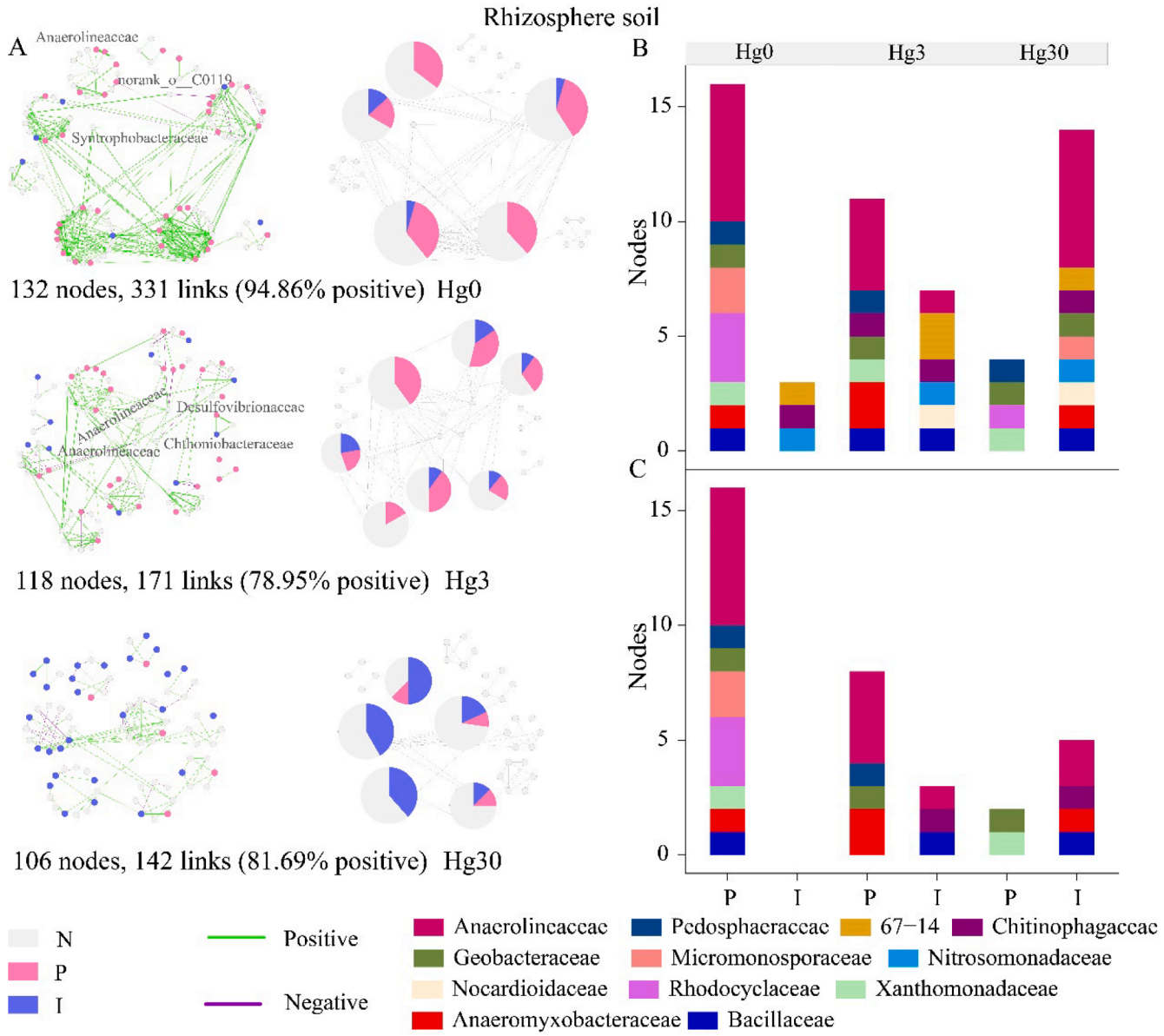


Fig. 4. Highly connected modules within RS networks along an Hg gradient (A) and the number of nodes of the top bacteriaceae in different putative MeHg-associated taxa in the network (B) and highly connected modules (C). Colours of nodes indicate different putative Hg-associated taxa, pie charts represent the composition of modules with >8 nodes, and the nodes labeled with bacteriaceae are module hubs and connectors, among which nodes in the middle of the module are module hubs, and other nodes are connectors.

Species related to *Syntrophobacter* are ubiquitous in wetlands and freshwater ecosystems and may be propionate-dependent sulfate reducers and Syntrophs in paddy fields (Podar et al., 2015). From an ecological perspective, these results imply that IRB are dominant methylators in rice RS (Liu et al., 2018), and sulfate reduction is the main driving factor for the formation and concentration of MeHg (Wu et al., 2020). The families, known as putative Hg methylators (Gilmour et al., 2013; Liu et al., 2018; Liu et al., 2019; Ma et al., 2019a; Parks et al., 2013; Podar et al., 2015), did not dominate BS communities (Fig. 4 and S7). The proportion of Hg-promoted and Hg-inhibited methylators in the network modules regulated MeHg/THg of BS in response to the Hg gradient, suggesting that MeHg production in BS was primarily driven by Hg-promoted and Hg-inhibited methylators, and that Anaerolineaceae, Xanthomonadaceae and Gemmatimonadaceae were the taxa associated with Hg methylation (Fig. 3), thus differing from those in RS.

Hg demethylation by microbes in RS differs from BS as well. The RS is still a hot spot for Hg demethylation by multiple mechanisms of MeHg degradation. With the increase of WHg, the Hg concentration of 30 mg kg^{-1} decreased the abundance of *merA* in RS (Fig. S14). The Bacillaceae, belonging to demethylators (Oregaard and Sørensen, 2007), was the dominant family of RS communities (Fig. S7), but had a low number of nodes in RS networks (Fig. 4), indicating that the degradation of MeHg in RS was largely driven by Hg-promoted and Hg-inhibited methylators. Actually, in addition to reductive demethylation, oxidative demethylation, another MeHg degradation mechanism (Marvin-DiPasquale and Oremland, 1998) was indeed observed to participate in the demethylation process of rice RS (Wu et al., 2020; Zhou et al., 2020). Hence, additional studies are warranted to further explore which mechanism dominates MeHg degradation and whether the two mechanisms known are interrelated in RS. In BS, the Bacillaceae was the dominant family of microbial communities (Fig. S7) and had a top number of nodes in networks (Fig. 3), which may dominate Hg demethylation. Microbes with Hg demethylation ability, such as Catenulisporaceae, Frankiaceae, Mycobacteriaceae and Thermomonosporaceae (Zhou et al., 2020), did

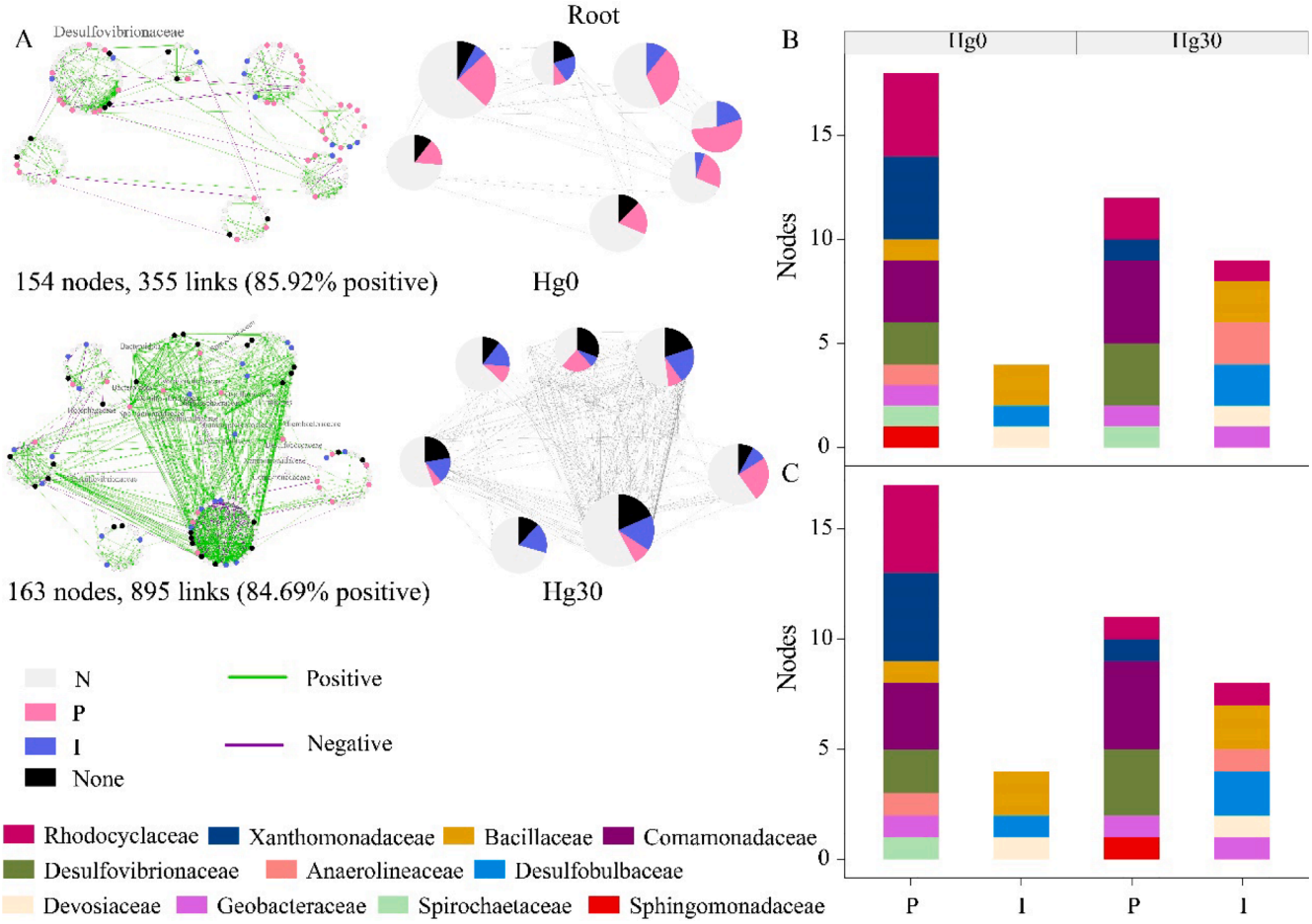


Fig. 5. Highly connected modules within root networks along an Hg gradient (A) and the number of nodes of the top bacteriaceae in different putative MeHg-associated taxa in the network (B) and highly connected modules (C). Colours of nodes indicate different putative Hg-associated taxa, pie charts represent the composition of modules with >8 nodes, and the nodes labeled with bacteriaceae are module hubs and connectors, among which nodes in the middle of the module are module hubs, and other nodes are connectors.

not assigned to the dominant family in paddy soil (Fig. S7) and had fewer nodes in soil networks (Figs. 3 and 4). The soil networks implied that Anaerolineaceae, 67-14, Chitinophagaceae, Micromonosporaceae, Ncardioidaceae and Bacillaceae were likely to be microbial taxa associated with MeHg degradation.

4.2. *Microbial mechanisms of MeHg production affected by rice development differ between BS and RS*

MeHg production is promoted by assemblages of RS communities regulated by rice plant development, but is inhibited in BS. The abundance of *merA* and *hgcA* genes in RS at the blooming stage was not different from that at the reviving stage (Fig. S14). However, module hubs of their networks were Nitrososphaeracea, Vicinamibacteraceae and Oxalobacteraceae belonging to Hg-promoted taxa despite the connector constituted Hg-inhibited taxa assigned to Gracilibacteraceae (Fig. S12). The assembly of the microbial communities in rice RS did not hinder MeHg production, reversely promoting Hg methylation to some extent. Probably, the accumulation of MeHg in rice is due to the rice plant itself, which promotes MeHg production by regulating the assemblages of microbial communities in RS and transfers the generated MeHg to the seeds (Guo et al., 2021; Rothenberg et al., 2014). The relationship between plants and soil microorganisms is primarily driven by chemical signaling molecules produced and exudated by the roots that attract beneficial microorganisms and influence microbial compositions in RS in order to improve plant performance (Bulgarelli et al., 2013). Among these substances, low molecular weight organic acids can desorb metal ions from the solid phase to form soluble metal-low molecular weight organic acid complexes and serve as bioavailable carbohydrates that improve the activity of soil microorganisms (Gunina et al., 2017; Wang and Mulligan, 2013). Therefore, BS microbial network varied in the present study with the RS network, in which the connector at the reviving stage and the module hub at blooming stage belong to Hg-inhibited taxa (Fig. S11). The difference between BS and RS networks was presumably mediated by detoxification mechanisms of BS microorganisms responding to Hg stress regulated by rice. Under heavy metal stress, microorganisms grow on root exudates and compete for ecological niches and resources (Thijs et al., 2017). Root exudates form a concentration gradient pattern in RS determined by their solubility and stability, with some organic matter spreading over longer distances (Bais et al., 2006). WHg ($R = 0.60$, $P < 0.001$) and MeHg ($R = 0.88$, $P < 0.001$) in BS and RS are positively correlated, suggesting that BS can be regarded as an Hg source of RS. Thereby, differences between BS and RS networks may also be due to different regulatory mechanisms under Hg stress mediated by the concentration gradient of rice root exudates in BS and RS. The processes by which root exudates regulate the assembly of BS and RS microorganisms and their assembly requires further studies.

4.3. *Microbes in roots play a role in the detoxification of Hg*

Endophytes could mediate the speciation and phytotoxicity of Hg,

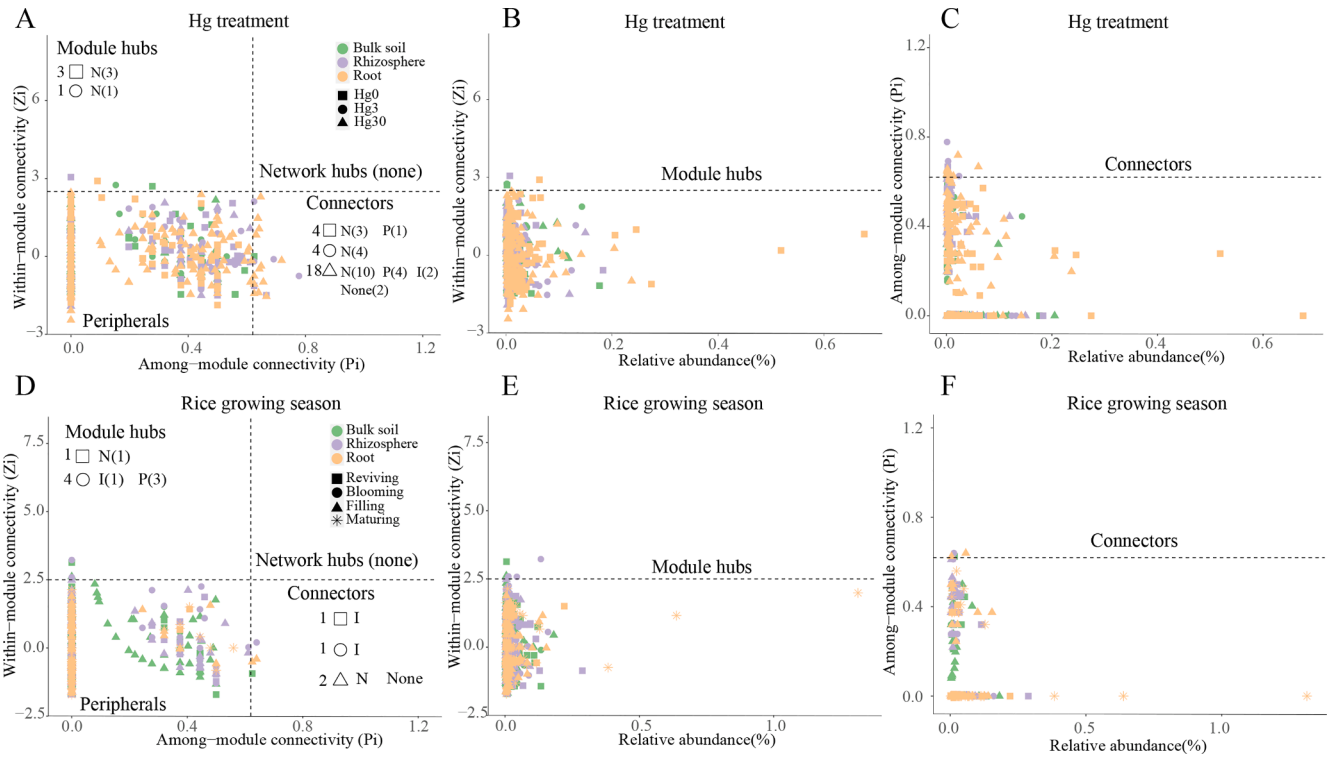


Fig. 6. Classification of nodes to identify putative keystone species within networks (A and B) and the relative abundance of module hubs and module connectors in networks (B, C, E and F). Each symbol represents an OTU from the 17 networks selected for detailed module analysis (Figs. 3, 4 and 5). Module hubs have Zi > 2.5, whereas connectors have Pi > 0.62. Detailed taxonomic information for module hubs and connectors is listed in Table S4.

improving plant resistance and growth, such as *Pseudomonas* sp. BacI38 and *Serratia marcescens* BacI56 enhancing Hg phytovolatilization (Mello et al., 2020; Yu et al., 2023). These improvements can be attributed to enhancement of the dissolution of essential metals and minerals through metabolites (Yu et al., 2023), and to redox reactions, chelation, polymerization, precipitation and coordination to alter the bioavailability of Hg and other heavy metals in farmland (Aghabaki Alooghareh et al., 2021). Inoculation with *Lindgomycetaceae* P87 and *Aspergillus* sp. A31 increased resistance to Hg in maize compared to inoculation with *Lindgomycetaceae* P87 (Pietro-Souza et al., 2020), and inoculation with endophytic bacteria (e.g. *Bacillus* sp. BacI34, *Klebsiella pneumoniae* BacI20, *Enterobacter* sp. BacI26) promoted the growth of maize in Hg-added soil (Sharma and Kumar, 2021). SRB are important for the response of rice roots to Hg. Changes in Hg concentrations had little effect on root microbial diversity and community structure, but the complexity of root networks increased significantly with increasing Hg concentration to 30 mg kg^{-1}. Under these conditions, the number of connectors was up to 18 (Fig. 6), with Desulfovibrionaceae appearing most frequently and constituting the module hub in root microbial networks at 0 mg kg^{-1} Hg (Table S4). Sulfur forms organic molecules in plants in the form of reduced organic mercaptan or organic sulfuric acid compounds through assimilation (Takahashi et al., 2011), such as cysteine, cystine, glutathione and phytochelation (PC). These sulfur-containing proteins or amino acids play an important role in the detoxification mechanism of plants in response to Hg stress. Glutathione synthesized by cysteine is a critical primary metabolite in plants. It forms the ascorbic acid- glutathione cycling system to resist oxidative damage caused by heavy metal stress (Davey et al., 2000; Nagajyoti et al., 2010). PCs in rice roots can effectively chelate Hg^{2+} to form Hg-PCs complex, thus retaining most of Hg^{2+} in rice roots (Krupp et al., 2009). Plants can enhance sulfur assimilation to improve the sulfur available for the detoxification of heavy metals by phytochelations and metallothioneins (Gupta et al., 2013). Therefore, sulfur deficiency can aggravate heavy metal stress in plants, while high sulfate supply can reduce heavy metal-induced toxicity (Dixit et al., 2016). Probably sulfate-reducing Desulfovibrionaceae participated in the detoxification of heavy metals in rice plants.

5. Conclusions

The present results show that paddy fields are hot spots for MeHg production, especially during flowering and filling stages. Most previous studies ignored the contribution of putative non-methylating microorganisms to MeHg generation and many microorganisms with unknown functions are also restricting our understanding of Hg methylation. In this study, we present network analyses from the MeHg production perspective, suggesting the role of these microorganisms in Hg methylation. Microbial processes for Hg methylation and demethylation were found to be different between RS and BS. MeHg production in RS was mainly dominated by putative Hg methylating microorganisms, but by other microorganisms in BS. The demethylation of MeHg may involve multiple mechanisms in RS and was dominated by the putative Hg-resistant microorganisms in BS. Moreover, sulfate-reducing Desulfovibrionaceae appeared to not merely drive MeHg generation in RS, but also be related to the Hg detoxification in rice roots. Since SRB are considered to be the main Hg-methylating microorganisms (Gilmour et al., 2013; Parks et al., 2013; Tang et al., 2020), future studies should focus on the association between sulfur metabolism in soil and rice plants as well as migration and transformation of Hg.

CRediT authorship contribution statement

Pan Guo: Conceptualization, Investigation, Methodology, Experiment, Data curation, Formal analysis, Writing – original draft. **Heinz Rennenberg:** Supervision, Funding acquisition, Formal analysis, Writing - review & editing. **Hongxia Du:** Funding acquisition, Formal analysis, Writing - review & editing. **Tao Wang:** Investigation, Writing - review & editing. **Lan Gao:** Investigation, Writing - review & editing.

Emmanouil Flemetakis: Formal analysis, Writing - review & editing. **Robert Hänsch:** Formal analysis, Writing - review & editing. **Ming Ma:** Conceptualization, Methodology, Resources, Data curation, Formal analysis, Funding acquisition, Validation, Writing - review & editing. **Dingyong Wang:** Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envint.2023.108066.

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